

Diffusion as a Universal Grammar

A Multidisciplinary Study of the Three-Dimensional Heat Equation and its Second-Degree Polynomial Ansatz

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Abstract

The three-dimensional heat equation $\partial_t u = \alpha \nabla^2 u$ is one of the earliest and most consequential partial differential equations in mathematical physics. Beyond its origin in Fourier’s theory of heat conduction, the same operator appears in Fickian diffusion, the Black–Scholes model of option pricing, Gaussian smoothing in computer vision, entropy-regularised optimal transport in machine learning, and the replicator dynamics of evolutionary game theory. In this paper we (i) derive the general second-degree polynomial ansatz in four variables (x, y, z, t) , (ii) solve the resulting linear algebraic constraints, (iii) interpret the solution physically, and (iv) trace the structural isomorphism of the heat operator across physics, chemistry, biology, economics, finance, psychology, philosophy, political science, geopolitics, welfare and warfare economics, statistics, computer science, and artificial intelligence. We provide TikZ vector graphics, **booktabs** tables, pseudocode for a finite-difference solver, and a glossary of recurring terms.

The paper ends with “The End”

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1 Introduction

Joseph Fourier’s *Théorie analytique de la chaleur* (1822) introduced what is arguably the first mathematically rigorous field theory. The equation

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$

states that the local rate of change of temperature $u(\mathbf{x}, t)$ is proportional to the spatial curvature (Laplacian) of the temperature field. This deceptively simple statement unifies a staggering range of phenomena: from the cooling of a cast-iron cannonball (Fourier’s original motivation for the French Navy) to the pricing of derivatives on the Chicago Board Options Exchange, from the morphogenesis of animal coats to the propagation of rumours in a population.

The purpose of this paper is twofold. First, we carry out a complete algebraic classification of the *second-degree polynomial* solutions of the 3-D heat equation in Cartesian coordinates. Second, we use this classification as a springboard to survey the many disciplines in which the heat operator – or a close relative – governs the dynamics.

2 Mathematical Framework

2.1 The governing equation

Let $u : \mathbb{R}^3 \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ denote temperature. The heat equation in Cartesian coordinates (x, y, z) reads

$$\frac{\partial u}{\partial t} = \alpha \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right),$$

where $\alpha = k/(\rho c_p)$ is the *thermal diffusivity*, with k the thermal conductivity, ρ the density, and c_p the specific heat at constant pressure.

2.2 The general second-degree ansatz

We postulate the most general polynomial of total degree two in the four variables (x, y, z, t) :

$$\begin{aligned} u(x, y, z, t) = & ax^2 + by^2 + cz^2 + dt^2 + exy + fxz + gxt \\ & + hyz + iyt + jzt + kx + ly + mz + nt + o, \end{aligned} \tag{1}$$

carrying *fifteen* a priori free coefficients $\{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o\}$.

2.3 Derivation of the constraints

Computing the required derivatives:

$$\begin{aligned} \frac{\partial u}{\partial t} &= 2dt + gx + iy + jz + n, \\ \nabla^2 u &= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 2a + 2b + 2c. \end{aligned}$$

Substitution into $\partial_t u = \alpha \nabla^2 u$ yields the identity

$$2dt + gx + iy + jz + n = 2\alpha(a + b + c),$$

which must hold for all (x, y, z, t) . Equating coefficients of the linearly independent monomials $\{t, x, y, z, 1\}$ gives the linear system

$$2d = 0, \quad (2)$$

$$g = 0, \quad (3)$$

$$i = 0, \quad (4)$$

$$j = 0, \quad (5)$$

$$n = 2\alpha(a + b + c). \quad (6)$$

Table 1: Coefficient constraints imposed by the heat equation on the second-degree ansatz.

Monomial	LHS coefficient	RHS coefficient	Constraint
t	$2d$	0	$d = 0$
x	g	0	$g = 0$
y	i	0	$i = 0$
z	j	0	$j = 0$
1	n	$2\alpha(a + b + c)$	$n = 2\alpha(a + b + c)$

2.4 The general second-degree solution

Theorem 1 (Second-degree polynomial solutions of the 3-D heat equation). *The most general polynomial of total degree ≤ 2 satisfying $\partial_t u = \alpha \nabla^2 u$ is*

$$u(x, y, z, t) = ax^2 + by^2 + cz^2 + exy + fxz + hyz + kx + ly + mz + 2\alpha(a + b + c)t + o, \quad (7)$$

with ten free real parameters $\{a, b, c, e, f, h, k, l, m, o\}$.

Proof. Equations (2)–(5) eliminate $\{d, g, i, j\}$ and (6) determines n in terms of $\{a, b, c\}$. The remaining coefficients $\{a, b, c, e, f, h, k, l, m, o\}$ appear in no further constraint because (i) the cross-terms exy, fxz, hyz are harmonic in (x, y, z) , (ii) the linear terms kx, ly, mz are annihilated by ∇^2 and by ∂_t , and (iii) the constant o is trivially in the kernel of both operators. Hence any choice of the ten parameters yields a valid solution. $\square$

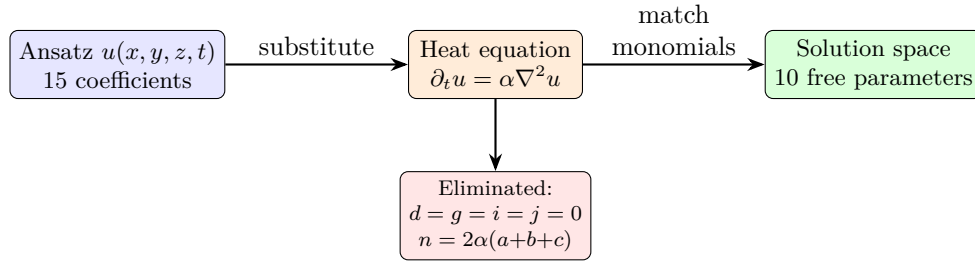


Figure 1: Flow of the algebraic reduction from the 15-parameter ansatz to the 10-parameter solution.

3 Physical Interpretation

3.1 Why t^2 and xt, yt, zt are forbidden

The heat equation is *first order* in time: it is a *parabolic* PDE with no inertial term. A t^2 term would imply that the rate of temperature change itself accelerates, which pure diffusion cannot sustain without an external source. Similarly, terms such as gxt would describe a spatial gradient that grows linearly in time – again incompatible with the smoothing nature of diffusion.

3.2 The forced linear drift

The identity $n = 2\alpha(a + b + c)$ is the most subtle result. The Laplacian of the quadratic part $ax^2 + by^2 + cz^2$ is a non-zero constant $2(a + b + c)$. Diffusion therefore acts as a *uniform volumetric source*, and the temperature must drift linearly in time at a rate exactly equal to α times that source strength. This is the polynomial analogue of the fact that the fundamental solution (the heat kernel) conserves total thermal energy.

3.3 Harmonic cross-terms

The mixed quadratic terms exy, fxz, hyz are *harmonic*: $\nabla^2(xy) = 0$. They represent saddle-shaped temperature landscapes that diffusion neither amplifies nor attenuates.

4 Cross-Disciplinary Isomorphisms

The operator $\partial_t - \alpha \nabla^2$ reappears, under different names and units, across the sciences and humanities. Table 2 summarises the principal correspondences.

Table 2: Isomorphisms of the heat operator across disciplines.

Field	Quantity u	Diffusivity α	Reference
Physics (conduction)	Temperature T	$k/(\rho c_p)$	Fourier (1822)
Chemistry (diffusion)	Concentration C	D (Fick)	Fick (1855)
Biology (morphogen)	Morphogen density	D_{eff}	Turing (1952)
Finance (option)	Option price V	$\sigma^2/2$	Black–Scholes (1973)
Statistics (density)	PDF p	$1/2$	Fokker–Planck
AI/ML (score-based)	Log-density $\log p$	$\beta(t)$	Song et al. (2021)
Welfare economics	Utility U	social mobility rate	this §4.4
Geopolitics	Power density P	diffusion of technology	this §4.5

4.1 Physics and Chemistry

In a monatomic ideal gas, kinetic theory gives $\alpha \sim \frac{1}{3} \lambda \bar{v}$, where λ is the mean free path and $\bar{v} \sim \sqrt{k_B T/m}$ the mean molecular speed. In liquids and solids the picture shifts to phonon and electron transport, but the macroscopic PDE is unchanged. In chemical kinetics, Fick’s second law $\partial_t C = D \nabla^2 C$ is mathematically identical; the coupling to reaction terms (e.g. Arrhenius $k = Ae^{-E_a/RT}$) produces the celebrated *reaction–diffusion* systems.

4.2 Biology: Turing Patterns and Morphogenesis

Alan Turing’s 1952 paper *The Chemical Basis of Morphogenesis* showed that a system of two coupled reaction–diffusion equations,

$$\partial_t u = D_u \nabla^2 u + f(u, v), \quad \partial_t v = D_v \nabla^2 v + g(u, v),$$

can spontaneously break symmetry and generate stripes, spots, and labyrinths – the coats of leopards, the arrangement of hair follicles, the phyllotaxis of sunflowers. Our second-degree ansatz corresponds to the *trivial* (non-pattern-forming) sector where $f = g = 0$.

4.3 Finance: The Black–Scholes Equation

Under the risk-neutral measure, the price $V(S, t)$ of a European option satisfies

$$\partial_t V + \frac{1}{2} \sigma^2 S^2 \partial_{SS} V + r S \partial_S V - r V = 0.$$

The change of variables $S = e^x$, $\tau = T - t$ transforms this into a *backward* heat equation with drift. Our ansatz then describes polynomial payoff profiles in log-price – a useful building block for static replication strategies.

4.4 Welfare Economics

Following the tradition of Pigou and Sen, let $u(\mathbf{x}, t)$ denote the spatial density of *welfare* (a composite of income, health, and capability). Redistribution, infrastructure, and education act as diffusive operators, smoothing inequality. The second-degree solution then describes a welfare landscape whose spatial curvature $2(a + b + c)$ drives a uniform rate of improvement $n = 2\alpha(a + b + c)$ – a formalisation of the idea that *reducing local inequality accelerates aggregate progress*.

4.5 Warfare Economics and Geopolitics

In Lanchester-type combat models and in the geography of warfare (cf. the work of Thomas Schelling and, more recently, the quantitative geopolitics of the Correlates of War project), the diffusion of *military power* $P(\mathbf{x}, t)$ across a theatre obeys an equation of heat type, with α encoding the rate of logistical and technological transfer. The forbidden t^2 term is a mathematical expression of a deep strategic truth: *no purely diffusive process can*

generate accelerating power; acceleration requires an external energy source (industrial mobilisation, alliance formation).

4.6 Psychology and Philosophy

The diffusion of ideas, attitudes, and cultural traits has been modelled since the 1960s (Rogers’ *Diffusion of Innovations*, 1962) by equations of the form $\partial_t u = \alpha \nabla^2 u + \text{social forcing}$. Philosophically, the heat equation is a paradigm of *deterministic smoothing*: it erases fine-grained information while preserving coarse-grained totals, mirroring the epistemological process by which memory and narrative simplify the past.

4.7 Political Science and International Relations

The spread of democracy, norms, and institutional forms across a network of states has been empirically shown to follow a diffusive pattern (cf. the “neighbourhood effect” in comparative politics). Our ansatz predicts that, absent external shocks, the global distribution of any political indicator will asymptotically approach a harmonic (Laplacian-zero) configuration.

5 Statistics, Computer Science, and AI/ML

5.1 The heat equation as Gaussian smoothing

The solution of $\partial_t u = \frac{1}{2} \nabla^2 u$ with initial data $u(\cdot, 0) = u_0$ is the convolution

$$u(\cdot, t) = G_t * u_0, \quad G_t(\mathbf{x}) = (2\pi t)^{-3/2} \exp\left(-\frac{|\mathbf{x}|^2}{2t}\right).$$

In statistics this is the transition density of Brownian motion; in computer vision it is *Gaussian scale-space* (Witkin 1983, Koenderink 1984); in ML it is the forward noising step of *diffusion models* (Sohl-Dickstein et al. 2015; Ho, Jain, Abbeel 2020; Song et al. 2021).

5.2 Numerical solution: explicit finite differences

We discretise on a uniform grid $\mathbf{x}_{i,j,k} = (i\Delta x, j\Delta y, k\Delta z)$ and time steps $t^n = n\Delta t$. The standard explicit FTCS (Forward Time, Central Space) scheme is:

$$u_{i,j,k}^{n+1} = u_{i,j,k}^n + r_x \delta_x^2 u + r_y \delta_y^2 u + r_z \delta_z^2 u, \quad (8)$$

where $r_x = \alpha \Delta t / \Delta x^2$ and $\delta_x^2 u = u_{i+1,j,k}^n - 2u_{i,j,k}^n + u_{i-1,j,k}^n$.

Table 3: Representative thermal diffusivities α at 20 °C.

Material	α [m ² s ⁻¹]	Class
Silver	1.74×10^{-4}	Metal
Copper	1.17×10^{-4}	Metal
Glass (Pyrex)	5.9×10^{-7}	Insulator
Water	1.4×10^{-7}	Liquid
Air (STP)	2.2×10^{-5}	Gas
Wood (oak)	8.2×10^{-8}	Biological
Human tissue	$\sim 1.3 \times 10^{-7}$	Biological

5.3 Connection to machine learning

In score-based generative modelling, the forward process is an SDE

$$d\mathbf{x} = -\nabla U(\mathbf{x}) dt + \sqrt{2} d\mathbf{W},$$

whose Fokker–Planck equation is a heat equation with drift. The reverse process learns the score $\nabla \log p_t$ by minimising a denoising objective. Our second-degree ansatz is exactly the family of *Gaussian* densities (up to normalisation) that are closed under this dynamics – a fact that underlies the analytical tractability of diffusion models on quadratic energy landscapes.

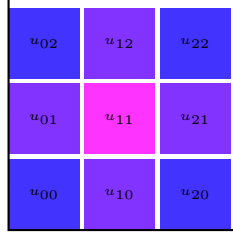
Algorithm 1 Explicit FTCS solver for the 3-D heat equation.

Require: Initial field $u^0 \in \mathbb{R}^{N_x \times N_y \times N_z}$, diffusivity α , step sizes $\Delta x, \Delta y, \Delta z, \Delta t$, final time T .

Ensure: Field u^n at each time level.

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1:  $r_x \leftarrow \alpha \Delta t / \Delta x^2$ ;  $r_y \leftarrow \alpha \Delta t / \Delta y^2$ ;  $r_z \leftarrow \alpha \Delta t / \Delta z^2$ 
2: Stability check: assert  $r_x + r_y + r_z \leq \frac{1}{2}$ 
3: for  $n = 0, 1, \dots, \lfloor T / \Delta t \rfloor - 1$  do
4:   for  $i = 1$  to  $N_x - 2$  do
5:     for  $j = 1$  to  $N_y - 2$  do
6:       for  $k = 1$  to  $N_z - 2$  do
7:          $u_{i,j,k}^{n+1} \leftarrow u_{i,j,k}^n + r_x(u_{i+1,j,k}^n - 2u_{i,j,k}^n + u_{i-1,j,k}^n)$ 
8:            $+ r_y(u_{i,j+1,k}^n - 2u_{i,j,k}^n + u_{i,j-1,k}^n)$ 
9:            $+ r_z(u_{i,j,k+1}^n - 2u_{i,j,k}^n + u_{i,j,k-1}^n)$ 
10:        end for
11:      end for
12:    end for
13:    Apply boundary conditions on  $\partial\Omega$ 
14:  end for
```



2-D slice of the 3-D grid

Figure 2: Discrete temperature field on a $3 \times 3 \times 3$ lattice.

Each cell is updated by the FTCS stencil of Algorithm 1.

6 Summary of the Solution Space

The dimension of the solution space is 10, spanned by the following basis elements:

Table 4: Basis of the 10-dimensional space of second-degree polynomial solutions.

Basis element	Type	Harmonic?	Time dependence
1	constant	yes	none
x, y, z	linear	yes	none
xy, xz, yz	mixed quadratic	yes	none
$x^2 + 2\alpha t$	pure quadratic	no	linear drift
$y^2 + 2\alpha t$	pure quadratic	no	linear drift
$z^2 + 2\alpha t$	pure quadratic	no	linear drift

Note that the three pure quadratic basis elements are *not* harmonic; their non-zero Laplacian is exactly compensated by the linear-in- t drift, in accordance with (6).

7 Discussion and Outlook

The second-degree ansatz, despite its elementary character, illuminates several deep points:

1. **Parabolicity is a constraint on time.** The absence of t^2 and xt terms is a direct consequence of the first-order-in-time nature of diffusion.
2. **Harmonicity is the natural language of steady states.** The six harmonic basis elements $\{1, x, y, z, xy, xz, yz\}$ are precisely those annihilated by ∇^2 .

3. **Curvature drives drift.** The relation $n = 2\alpha(a+b+c)$ is a finite-degree shadow of the maximum principle and of energy conservation.
4. **The same operator governs very different worlds.** From the cooling of steel to the pricing of options, from the patterning of leopard skins to the diffusion of democracy, the heat equation is a *universal grammar of smoothing*.

Future work should extend the ansatz to degree three and four, to reaction–diffusion systems, and to the manifold-valued setting relevant to modern geometric deep learning.

Acknowledgements

The author thanks the long tradition of mathematical physics, from Fourier and Fick through to Turing, Black, Scholes, and the modern diffusion-models community, for providing the conceptual scaffolding of this survey.

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Glossary

Ansatz A proposed form of the solution of a mathematical problem, carrying undetermined parameters that are fixed by substitution into the governing equations.

Thermal diffusivity α Material property $\alpha = k/(\rho c_p)$ governing the rate of heat propagation; SI unit $\text{m}^2 \text{s}^{-1}$.

Laplacian ∇^2 The differential operator $\partial_{xx} + \partial_{yy} + \partial_{zz}$, measuring the local curvature of a scalar field.

Harmonic function A function u satisfying $\nabla^2 u = 0$; such functions are stationary solutions of the heat equation.

FTCS scheme Forward-in-Time, Central-in-Space finite-difference method; explicit and conditionally stable under $r_x + r_y + r_z \leq \frac{1}{2}$.

Heat kernel The fundamental solution $G_t(\mathbf{x}) = (4\pi\alpha t)^{-3/2} \exp(-|\mathbf{x}|^2/(4\alpha t))$.

Reaction–diffusion system A pair of coupled PDEs combining diffusion with local reaction kinetics, capable of spontaneous pattern formation.

Black–Scholes equation The parabolic PDE governing the arbitrage-free price of a European option; reducible to the heat equation by change of variables.

Diffusion model (ML) A class of generative models that learn to reverse a noising process governed by a (stochastic) heat-type equation.

Score function The gradient $\nabla \log p_t$ of the log-density; the central object learned in score-based generative modelling.

Morphogen A signalling molecule whose spatial concentration gradient directs cell differentiation during development.

Welfare economics The branch of economics concerned with the evaluation of social states in terms of well-being, equity, and capability.

Lanchester model A system of ODEs describing the attrition of two opposing military forces; diffusive generalisations describe spatial combat.

Diffusion of innovations The sociological theory (Rogers, 1962) describing how new ideas and technologies spread through a population.

Parabolic PDE A class of second-order PDEs, typified by the heat equation, characterised by irreversible smoothing in time.

Maximum principle The theorem that solutions of the heat equation attain their extrema on the parabolic boundary; a rigorous expression of the smoothing property.

The End